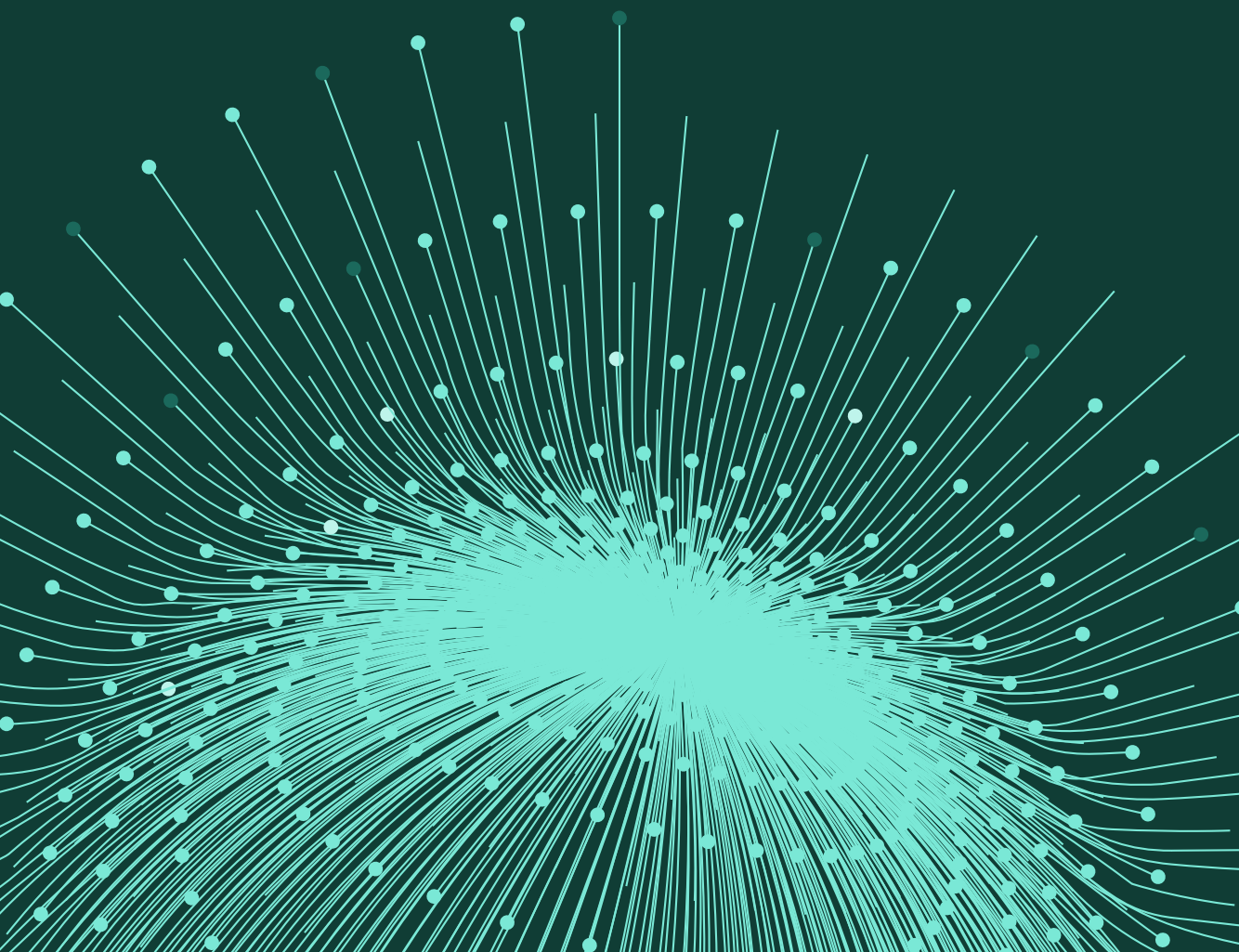


GUIDE



Authorised push payment fraud detection for PSPs

A behavioural playbook



Executive summary

Authorised push payment (APP) fraud is the fastest-growing payment crime in the UK. In H1 2025, losses reached £257.5 million - 41% of all UK fraud losses, up from 38% the prior year.¹

Under the PSR's reimbursement framework, every PSP in the payment chain now absorbs a share.

In APP fraud, the victim initiates the transfer on their own recognized device, with genuine authentication. Every conventional control passes the transaction. The fraud happens before the payment screen, in the session, in the victim's behaviour. That's where detection has to operate.

This guide covers:

- Why APP fraud breaks conventional transaction-monitoring models.
- All seven major APP scam typologies and their behavioural fingerprints.
- How to detect fraud on both the sending and receiving sides of the payment chain.
- How AI voice cloning is outpacing classic impersonation detection.
- What Confirmation of Payee does and doesn't solve.
- How the UK PSR's reimbursement framework changes the economics for every PSP in the chain.
- How Sardine's layered detection covers the full fraud lifecycle - from session to settlement.

The scale of the problem

UK APP fraud losses rose 12% year-over-year in H1 2025, reaching £257.5 million, 41% of all UK fraud losses. That share has grown each year while other fraud categories plateaued or declined.

APP fraud scales because it exploits a structural gap: the victim authorizes and executes the payment themselves. Card fraud uses a stolen number to fund a transaction the account holder never touched. APP fraud requires the victim to initiate the payment, believing the recipient is legitimate. Conventional controls miss it because the authorization is genuine. Losses are expensive to absorb because reimbursement now falls on the PSP.

Scams globally are now a major criminal economy, with some industry estimates putting losses at \$75 billion and projecting 40% annual growth that could push losses to \$10 trillion by 2030.² For UK payment firms, the detection gap is no longer just a technical issue; with mandatory reimbursement for many APP scams now in force, it has become a direct financial liability.

Why APP fraud breaks conventional detection

Most payment fraud leaves a technical anomaly. Card fraud uses a compromised number from an unrecognized device or location. Unauthorised ACH transactions happen without the account holder's involvement. Conventional fraud detection looks for those anomalies: an authorization that shouldn't have happened.

APP fraud inverts that model. The account holder performs the transaction from their own device, on their own network, within normal hours. The amount may be elevated but plausible. The payee is new, but first-time payees are routine. A rules engine checking velocity, geolocation, card-not-present signals, or device risk sees nothing wrong.

The transaction layer misses everything happening around the payment: a fraudster coaching the victim by phone through each screen; a remote desktop tool running while a scammer navigates the account; a victim pausing on confirmation screens, rereading warnings they'd normally dismiss in seconds, typing dictated account details they'd normally recall from memory.

Those signals live in the session, and that's where APP fraud detection has to operate.

Seven typologies and their behavioural fingerprints

Each APP typology produces distinct behavioural patterns that session data can surface before a payment clears.

TYPOLOGY	MECHANISM	TYPICAL CHANNEL	AVG LOSS (UK)
Investment / Pig Butchering	Fake trading platforms; crypto push	Social media WhatsApp	£19,000+
Impersonation	Spoofed bank/police/HMRC call	Phone (spoofed CLI)	£5,000–£20,000
Romance	Trust built over weeks; money extracted	Instagram Dating apps	£8,000–£50,000
Purchase	Fake listings; goods never delivered	Facebook Marketplace Web	£650 on average
Invoice / BEC	Redirected supplier payment	Email / Corporate	£50,000+ (B2B)
Advance fee	Upfront "fee" to release larger sum	SMS Email	£1,500 on average
Job or task	Deposit crypto to "unlock" earnings	SMS Telegram	\$7,000 on average

Investment scams / pig butchering

Estimated average loss: £19,000+

/1

Fraudsters pose as investment advisors or platforms, typically using cryptocurrency or precious metals as lures, and extract large one-time transfers. Victims exhibit deliberate, research-like session behaviour: careful navigation, returns to earlier screens, extended dwell times. The key pattern is pronounced hesitation on confirmation screens, then a sudden decision to complete a high-value payment to a first-time payee.

In pig butchering variants, fraudsters build trust over weeks, extract multiple escalating payments, then disappear. Payments each run 20-30% larger than the last. Systems that treat established payees as safe can be gamed by fraudsters who deliberately season the relationship with small early payments.

Impersonation scams

Estimated average loss: £5,000–£20,000

/2

Fraudsters pose as banks, government agencies, HMRC, or law enforcement, convincing victims their funds are at risk and directing them to transfer money to a “safe” account the fraudster controls. These produce the most pronounced session anomalies of any typology. Victims move quickly and anxiously, typically on a simultaneous phone call. Remote access tools (RATs), including AnyDesk, TeamViewer, and QuickSupport, are frequently installed at the fraudster’s direction. Hesitation on warning screens is pronounced as victims struggle to reconcile the fraud warning with the authoritative voice on the line.

When a RAT is detected during a payment session, present no in-session challenge. The fraudster is watching the screen. Out-of-band verification is the only safe response.

Romance scams

Estimated average loss: £8,000–£50,000

/3

Fraudsters build relationships with victims over weeks or months through social media or dating platforms, then request funds for fabricated emergencies or investment opportunities. Payments often span multiple sessions before the first transfer occurs. Session behaviour trends toward familiarity and deliberateness; the victim moves with apparent confidence because they believe they know the recipient.

Romance scams create a specific detection challenge: sophisticated operations make three to five small payments early in the campaign to season the payee relationship and trigger false-positive suppression rules on the large extraction payment. Systems should not suppress risk flags when the current payment exceeds the payee's prior average by more than 50%.

Purchase scams

Estimated average loss: £650

/4

Victims pay for goods or services the fraudster never intends to deliver, most commonly on Facebook Marketplace, eBay, Gumtree, or similar classified platforms. Purchase scams generate the highest transaction count of any APP type. Individual losses are lower; collectively they represent a material share of total APP losses. Session behaviour is typically fast and transactional. Key signals are the combination of first-time payee status, round amounts (multiples of £50), and payment reference text containing marketplace keywords.

Invoice / business email compromise (BEC)

Estimated average loss: £50,000+ (B2B)

/5

Fraudsters intercept or spoof supplier communications, redirecting payment to a fraudster-controlled account. The victim believes they are paying a legitimate, known vendor. The distinguishing signal is a new beneficiary account that shares name similarity with an existing payee in the customer's history: same company name, different

account number. CoP may return a Close Match rather than No Match, a weaker signal than the substitution warrants.

Advance fee scams

Estimated average loss: £1,500

/6

Fraudsters request an upfront payment, framed as a fee, tax, processing charge, or deposit, to release a larger sum that doesn't exist. Payment reference text containing words like "fee," "tax," "processing," or "release" is the primary distinguishing signal. Without reference text analysis, rule-level detection alone cannot separate advance fee payments from routine low-value transactions.

Job / task scams

Estimated average loss: £5,000

/7

The fastest-growing APP typology by volume, particularly among younger victims. Fraudsters recruit via Telegram, SMS, or social media to complete "tasks" on a platform, promising commissions, then ask victims to make small-to-moderate crypto deposits to "unlock" their earnings, which never materialize. The defining pattern is repeated, rapid, small-to-moderate crypto deposits to multiple destinations within a short window, distinct from the large single-payment pattern of investment scams.

Behavioural signals: What to look for

Capturing behavioural signals in real time during the payment session closes the detection gap. Weak signals accumulate into risk bands; no single weak signal should block a payment in isolation. A single very-high-confidence indicator, like an active RAT, can reach Critical on its own.

Remote access tool (RAT) detection

Screen-sharing or remote desktop software running during a payment session is the most specific APP scam signal available. AnyDesk, TeamViewer, QuickSupport, Zoho Assist, ScreenConnect, Chrome Remote Desktop, and VNC clients, when active at the point of payment, carry very high diagnostic confidence. In impersonation and tech-support scams, the fraudster watches and controls the session directly.

Active call detection

A payment made while the account holder is on an active phone call is a risk signal specific to APP fraud. Active call status combined with a new payee and elevated amount points strongly to a coerced payment. Impersonation and investment scams typically keep the fraudster on the line through the entire payment process.

Session timing anomalies

Coached victims pause at screens that normally take seconds. Extended dwell time on confirmation or review screens, where fraud warnings appear, indicates a victim rereading under instruction. Once a behavioural baseline is established, these patterns distinguish coached sessions from genuine ones. An impersonation victim who moves quickly through most of the flow then stalls at the confirmation page shows a different signature from a legitimate sender who pauses at the same point.

Copy-paste behaviour on payment fields

Victims given account details verbally or via message copy and paste sort codes and account numbers rather than typing them. Copy-paste activity on payment detail fields, in a session that otherwise shows natural typing, is a strong signal, particularly when combined with other session indicators.

Typing cadence consistent with dictation

When a fraudster dictates information over the phone, keystroke timing patterns differ from natural typing. Irregular rhythm, extended pauses between character groups, and correction patterns that match dictation indicate external coaching. Behavioural biometrics surface this signal without any knowledge of the payment content.

Hesitation on warning screens

Genuine users typically dismiss fraud warning dialogs in under three seconds. Victims often pause, sometimes navigating back and forth multiple times. Extended dwell or repeated navigation on warning pages is a direct diagnostic signal at the exact point where the fraudster's instructions conflict with the on-screen warning. Quick dismissal in under three seconds is also a weak risk signal, suggesting the victim is following instructions rather than reading.

Graduated intervention: Matching friction to risk

APP fraud intervention needs to be proportionate. Too little friction fails victims. Too much destroys customer experience, generates Consumer Duty complaints, and harms legitimate customers who are unnecessarily blocked.

A four-band graduated model builds risk scores from multiple signals in combination. No single weak signal drives a block.

BAND	SCORE RANGE	ACTION
Low	0–299	Proceed. Log for retraining.
Medium	300–599	Contextual scam-type warning + cognitive confirmation step.
High	600–799	Multi-step challenge questions + 24-hour cooling-off option + "Speak to fraud team" option.
Critical	800+	Suspend payment. Out-of-band verification only. 72-hour hold authorised.

A lone new-payee flag or round-amount match stays in Low, below 300. Two moderate signals push into Medium, triggering a contextual warning, not a block. Strong signal combinations reach High, presenting a multi-step challenge. Multiple strong signals or a single very-high-confidence indicator, such as an active RAT, reach Critical and suspend the payment.

When a RAT is active, the fraudster watches the victim’s screen. An in-session challenge is counterproductive; the fraudster will coach the victim through it. Suspend the payment, notify the customer via a separate channel, and have a fraud specialist call the registered number. No in-session challenge is trustworthy when a RAT is running.

Warning messages must be tailored to the detected scam type. An investment scam victim should see: “Were you introduced to this investment by someone you met online or through a message?” A romance scam victim: “Have you met this person in real life, or only online?” Generic boilerplate that doesn’t disrupt the fraudster’s script does almost nothing.

Consumers aged 70 and above, those with disclosed vulnerabilities, and those with prior APP fraud victimization in the past 24 months should face lower intervention thresholds: Medium activating earlier, challenge journeys using simpler language, and a prominent “Speak to someone now” option at every step. The PSR’s £85,000 reimbursement cap and the FCA’s Consumer Duty obligation both make differentiated protection a regulatory requirement.

The receiving side: Mule detection is half the problem

Most industry discussion on APP fraud focuses on the sending PSP. Proceeds, though, have to land somewhere. They land in accounts the fraudster controls: mule accounts opened with synthetic or stolen identities to receive and layer illicit funds.

Under the UK PSR's reimbursement framework, receiving PSPs share liability 50/50 with the sending firm, which changes the economics for every institution that receives payments.

Onboarding controls

Device intelligence during account opening surfaces indicators of synthetic or fraudulent identity at the point of application:

- **Factory-reset devices used immediately for account opening.** A device fully reset and then used to open a financial account is strongly associated with mule recruitment. Legitimate new customers overwhelmingly use established devices.
- **Emulators and virtual machines presenting as mobile devices.** Automated account creation operations frequently use emulation to scale. Device intelligence that detects emulated environments at onboarding blocks this vector before accounts activate.
- **Behavioural anomalies during the KYC flow.** Copy-pasting all identity fields, navigation speed inconsistent with genuine applicant behaviour, or session characteristics suggesting automation or external coaching are detectable signals.

Post-activation behavioural monitoring

Once a mule account is active, the following patterns are associated with fraudulent receiving activity:

- A dormant or low-activity account suddenly receiving multiple high-value inbound payments
- Immediate outbound transfers or cash withdrawals following inbound receipt
- Transaction timing that correlates with payments flagged on the sending side or with known fraud campaign windows
- Receiving patterns inconsistent with the account holder's stated purpose or expected activity profile

RUSI 2025 research on money mule networks found that approximately 20% of funds received by mule accounts exit via debit card activity rather than onward Faster Payments transfers. Standard mule detection focused on payment-to-payment velocity misses this exit route. Receiving institutions with visibility into card-present and ATM transactions should monitor for high inbound credit followed by debit card or ATM activity exceeding 50% of the credited amount within 24 hours, a pattern pure payment monitoring misses.

The first-victim problem

Consortium-based beneficiary risk data, signals about which receiving accounts other institutions have reported for fraud, stops compounding losses at known scam destinations. These signals only exist after the scam has claimed at least one victim at that account.

Consortium data limits the blast radius after a scam destination is flagged. It cannot detect a novel scam account on first use. Sending-side behavioural detection, covering behavioural biometrics, session signals, and new-payee analysis, is the primary control for first-victim prevention. Treating consortium signals as a substitute for behavioural and transactional detection creates a detection gap. Sophisticated fraudsters exploit it by routing initial payments through accounts not yet flagged.

Beneficiary risk signals

Receiving institutions carry very different risk profiles. Fintech platforms, neobanks, cryptocurrency exchanges, and forex platforms have historically maintained lower fraud controls than tier-1 and tier-2 banks, making them preferred mule destinations. Incorporating beneficiary bank risk into payment-time scoring, alongside beneficiary account age, number of unique senders previously associated with the account, and known scam complaint rates, improves sending-side detection without needing direct access to the receiving institution's data.

The emerging threat: AI voice and deepfake impersonation

APP scam detection was built for a world where impersonation required a real fraudster making a real call. Voice-cloning tools have changed that.

Fraud-as-a-Service platforms now offer voice impersonation for under \$100 per month. Fraudsters call victims from spoofed genuine numbers using a cloned voice indistinguishable from their bank's fraud team, with no RAT, no real person on the line, and no behavioural artifacts for conventional detection to catch.

Classic impersonation detection, concurrent call plus RAT plus new payee, catches less of this vector each month. The fraudster may call from a spoofed genuine number, with a cloned voice, and coach the victim through a legitimate-appearing session.

Telco metadata offers a partial answer. Call duration, originating number history, and repeat call patterns from numbers associated with known fraud campaigns can flag suspicious inbound calls before the victim reaches the payment screen. Real-time audio analysis in contact center interactions can detect AI-generated voices in calls to the fraud team.

The industry is catching up, but this gap isn't closed.. Mastercard's 2024 RAG-enabled voice scam detection achieved a 300% improvement in detection rates for this vector, demonstrating that the capability is achievable, though not yet standard. Payment firms should build AI voice impersonation into their APP fraud risk assessments now.

Confirmation of payee: Necessary, not sufficient

The UK's Confirmation of Payee (CoP) system runs a name-matching check before a payment clears. When the name the payer enters doesn't match the name on the receiving account, the payer receives a warning. Firms should incorporate CoP results into the broader risk score rather than treating them as a standalone pass/fail check: a Close Match adds risk weight; No Match or Account Not Found are strong signals.

CoP has real limits:

- Fraudsters increasingly register mule accounts under matching names. CoP confirms the match, reinforcing the victim's trust.
- In romance and investment scams, victims typically know, or believe they know, the name on the receiving account. CoP match confirmation reinforces that belief.
- In invoice/BEC scenarios, the fraudster provides a new account with a name similar but not identical to the legitimate vendor. CoP may return a Close Match rather than No Match, a weaker signal than the substitution warrants.
- CoP operates at the payment message level and has no visibility into the session behaviour that distinguishes a coerced sender from a voluntary one.

CoP reduces APP fraud at the margins - a supplementary control worth having and worth scoring, but one that leaves the behavioural detection gap untouched.

The regulatory shift: APP fraud as a balance sheet issue

The UK PSR's mandatory reimbursement framework (PS24/5, effective October 2024) requires both sending and receiving PSPs to reimburse APP fraud victims, with liability split 50/50 and a per-claim cap of £85,000. Before this framework, losses sat with victims. Now every PSP in the payment chain absorbs a share.

The £85,000 cap is contested. Consumer advocacy groups argue it is insufficient given that many investment and romance scams involve six-figure losses. The cap may increase, be restructured, or disappear, but APP fraud has become a permanent financial liability for payment firms.

The PSR framework also creates explainability obligations. Institutions must explain their reimbursement decisions: what interventions were presented, what the customer confirmed, and what the risk score was at the time of payment. Automated evidence packaging, capturing the full session context, risk score, feature breakdown, intervention history, and customer responses, is a compliance requirement.

The PSR framework allows institutions to withhold reimbursement in gross negligence cases, where a customer explicitly confirmed intent after a High or Critical intervention, and the customer was not in a vulnerable category. Analyst judgment alone creates FOS overturn risk. Automated assessment based on structured intervention evidence, whether the customer confirmed deliberately and whether they were flagged as vulnerable, is more consistent and more defensible.

The UK model is influencing international policy. For firms operating across jurisdictions, APP fraud detection is becoming a regulatory baseline, not a differentiator.

Sardine's approach: Layered detection across the full fraud chain

APP fraud requires controls at multiple points in the payment lifecycle. Sardine combines real-time behavioural detection, identity fraud prevention, and cross-network intelligence in a single platform.

Device intelligence and behavioural biometrics (DIBB)

Sardine's Device Intelligence + Behavioural Biometrics (DIBB) layer captures how a user interacts with an application, including typing cadence, hesitation patterns, copy-paste behaviour, swipe and scroll dynamics, remote access tool presence, and active call status, alongside device fingerprinting and session context. In APP fraud scenarios, DIBB surfaces coached or coerced sessions that transaction-level controls miss.

DIBB operates in real time, producing risk signals at the moment of payment that trigger graduated responses: additional authentication, a cooling-off delay, a first-time payee confirmation prompt, or escalation to manual review, all before the payment is authorised.

Real-time risk scoring at the point of payment

Sardine's risk engine scores payment sessions in real time using 4,800+ risk features, including behavioral signals, device context, session anomalies, first-time payee indicators, beneficiary account age, and beneficiary bank risk profile. The scoring architecture uses an additive (scoresum) model: every signal that fires contributes its score to a running total, the cumulative score determines the risk band, and the risk band determines the intervention.

Pre-built rule bundles cover all seven major APP typologies. Shadow-mode testing and rule backtesting allow firms to tune thresholds against historical data before deploying changes to production. Conditional workflows enable responses calibrated to risk level: moderate risk triggers a contextual warning,

high risk triggers a multi-step challenge, and Critical risk (a RAT detected, or multiple strong signals in combination) triggers payment suspension with out-of-band verification.

Identity fraud detection and mule prevention at onboarding

Sardine's onboarding capabilities detect synthetic and stolen identities at the point of account application, blocking mule infrastructure before it can receive APP fraud proceeds. KYC/KYB orchestration, document verification, and sanctions/PEP screening operate in a single workflow. An AI-assisted onboarding agent resolves name and date-of-birth mismatches to reduce false positives on legitimate applicants while maintaining high detection rates on fraudulent ones.

AML transaction monitoring and network graph analysis

Sardine's AML transaction monitoring and network graph tooling support investigation of connected entities after fraud surfaces, mapping shared devices, IP addresses, identity signals, and transaction flows to uncover fraud rings. One-click coordinated blocklisting of connected fraudulent accounts disrupts organized mule networks and removes the receiving infrastructure before it funds the next round of payments.

Consortium intelligence

APP fraud operates across institutions. Mule networks recruit at scale across multiple banks and platforms; fraud campaigns target multiple PSPs simultaneously. Sardine's consortium enables real-time data sharing across member institutions, surfacing receiving accounts flagged at other firms, identifying devices associated with fraud activity across the network, and providing consortium-level beneficiary risk signals that no single institution's data can replicate.

Consortium signals are most powerful for limiting blast radius after a scam destination is identified. Firms contributing to and consuming from the consortium gain visibility into beneficiary account fraud history that their own transaction data alone cannot surface.

Sponsor bank and BaaS coverage

For banks acting as sponsors to fintech programs, a common structure in UK e-money and EMI arrangements, PSR liability may cascade from the customer-

facing fintech back to the sponsor. Sardine's Sponsor Bank module allows sponsors to monitor and enforce fraud and compliance controls across all fintech programs and their customers in a single solution, giving sponsors visibility they would otherwise lack when they have no direct customer relationship.

GenAI-assisted case management and PSR evidence packaging

When a payment reaches the High or Critical threshold and a case opens, Sardine's GenAI copilot accelerates resolution: generating SAR and UAR narratives, pre-populating PSR claim fields with structured evidence from the session record, and surfacing the risk score, feature breakdown, intervention history, and behavioural session summary in a single analyst view. This keeps firms within the PSR's 5-day settlement window without assembling evidence manually across disconnected systems.

What comes next for APP fraud detection

Better transaction rules don't catch APP fraud. The transaction looks clean and the authorization is genuine. The signal that separates a fraud victim from a legitimate sender lives in the session: the hesitations, the copy-paste behaviour, the active phone call, the remote access tool running in the background. Capturing that signal requires session-level detection that most conventional fraud stacks lack.

The UK PSR's reimbursement framework has made APP fraud a balance sheet question for every PSP in the chain. Detection investment now competes directly with reimbursement liability on every firm's P&L.

Firms that build behavioural detection, mule prevention, and consortium-level beneficiary intelligence now will be positioned when the reimbursement cap rises, when a new jurisdiction introduces liability-shift rules, or when AI voice impersonation outruns today's detection.

Sardine addresses the full APP fraud chain in a single platform: behavioural signals at the point of payment, identity fraud prevention at onboarding, ongoing mule detection in receiving accounts, and cross-network intelligence from consortium data. Every stage of the fraud lifecycle has a control point.

Sources and data

- 1. <https://www.ukfinance.org.uk/news-and-insight/press-release/over-ps600-million-stolen-fraudsters-in-first-half-2025>
- 2. <https://time.com/6836703/pig-butcher-ing-scam-victim-loss-money-study-crypto/>
- UK Finance / Payment Systems Regulator APP scams reimbursement dashboard (H1 2025 data)
- PSR Mandatory Reimbursement Requirement PS24/5, effective October 2024
- RUSI 2025 money mule research (debit card exit route data)
- Forrester Research on behavioural risk scoring demand among European banks and PSPs
- Sardine APP Scam Detection PRD v1.0 (March 2026); APP Scam Operator Playbook v2.0 (April 2026)
- FTC 2024 data on job/task scam losses

About Sardine

Sardine is the AI risk platform for fraud, AML, and compliance. Sardine combines device intelligence, behavioural biometrics, real-time transaction monitoring, and consortium data to give financial institutions comprehensive protection against modern fraud, including APP scams, account takeover, synthetic identity fraud, and money mule networks. Sardine serves banks, fintechs, payment processors, and crypto platforms globally.

Learn more at sardine.ai